

Kasra Sinaei

Robotics Software Engineer | Physical AI Research Scientist | Authorized to work in the US

kasra.sinaei@gmail.com • linkedin.com/in/kasra-s-496843a2 • github.com/Kasra-sinaei • kassra-sinaei.github.io

SUMMARY

PhD candidate in Electrical Engineering working at the intersection of **Physical AI** and control theory. Industry and academic experience in robot learning, safety-critical robotic systems, and generative AI for robotics and control. Tested and deployed research concepts in production on humanoid, legged, mobile, and manipulator platforms.

EDUCATION

Pennsylvania State University – PhD, Electrical Engineering State College, PA | Expected May 2027

Dissertation: Safety Guarantees for Learning-Based Policies via Barrier Functions • GPA 3.90 / 4.00

Pennsylvania State University – MSc, Electrical Engineering

State College, PA | May 2024

Thesis: Model-Free Safety-Critical Position Control of Robots • GPA 3.84 / 4.00

University of Tehran – BSc, Mechanical Engineering

Tehran, Iran | Feb 2022

Capstone: Motion Planning and Control of a Wheeled-Legged Robot • GPA 17 / 20

WORK EXPERIENCE

Chewy – AI Research Scientist Intern

Boston, MA | Jun 2026 – Sep 2026

- Built a **LangGraph**-based agentic orchestration framework coordinating perception, task planning, and **vision-language-action (VLA)** policies for mobile manipulation in warehouse environments.
- Deployed the framework on a mobile manipulator for long-horizon, multi-step pick-and-place tasks, with semantic safety filtering in the control loop.

AlphaZ Inc. – Robotics Software Engineer Intern

Pasadena, CA | May 2025 – Aug 2025

- Developed **whole-body control** and **motion planning** for a humanoid platform with two 6-DoF arms on a differential-drive mobile base, running in real time on hardware.
- Delivered autonomous door opening – handle detection, approach, and manipulation – using a **Model Predictive Path Integral (MPPI)** planner; published at ASME MECC 2026.
- Built high-reliability **ROS 2** perception pipelines for state estimation and object detection, and integrated research code into the production navigation stack.

Penn State (CARL) – Graduate Research Assistant

State College, PA | Aug 2022 – Present

- Develop **safety guarantees** for learning-based control policies via **Control Barrier Functions (CBF)**, bridging classical control theory and foundation-model policies.
- Implemented model-free **safety filters** that require no analytical dynamics model, validated by **Lyapunov** and set-invariance analysis.
- Validated methods on hardware – Unitree Go1 quadruped and Quanser QArm manipulator – in addition to simulation.

University of Tehran (CAST) – Undergraduate Research Assistant

Tehran, Iran | Sep 2019 – Mar 2022

- Developed core software for a full-size **humanoid robot**: low-level drivers, **state estimation** fusing IMU and encoders, and closed-loop balance controllers.
- Designed gait pattern generators based on the **Divergent Component of Motion (DCM)** for traversing uneven terrain.

PUBLICATIONS

- [1] K. Sinaei and D. Ebeigbe, **Control barrier functions for adaptive stabilization of fully unknown underactuated nonlinear systems**, in *IEEE Conference on Decision and Control (CDC)*, Under Review, 2027.
- [2] K. Sinaei and D. Ebeigbe, **Safe adaptive control with vanishing conservativeness for robotic systems with unknown dynamics via barrier functions**, *IEEE Robotics and Automation Letters (RA-L)*, 2026.

- [3] K. Sinaei, K. Weerakoon, C. Bradley, S. A. Fakoorian, and D. Ebeigbe, **Motion planning for mobile manipulators navigating doorways via model predictive path integral**, in *ASME Modeling, Estimation and Control Conference (MECC)*, American Society of Mechanical Engineers, 2026.
- [4] K. Sinaei, H.-C. Wu, and D. Ebeigbe, **Safe vision language action model via barrier enhanced flow matching**, *IEEE Robotics and Automation Letters (RA-L)*, Under Review, 2026.
- [5] H.-C. Wu, X. Zhang, K. Sinaei, R. Abnavi, K. Weerakoon, C. Bradley, S. Fakoorian, J. Li, and D. Ebeigbe, **Strategic and reactive stream partitioning for path-efficient lidar-based obstacle avoidance**, in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2026.
- [6] K. Sinaei and D. Ebeigbe, **Safe robust control of nonlinear systems with uncertain regressor and parameter vector**, in *IEEE Conference on Control Technology and Applications (CCTA)*, 2025.
- [7] K. Sinaei, H.-C. Wu, and D. Ebeigbe, **Safety-critical position control of robots: A model-free approach**, in *American Control Conference (ACC)*, 2025.
- [8] P. Abdolhazehzad, A. Yousefi-Koma, A. Vedadi, K. Sinaei, B. Maleki, and M. Shafiee, **Online bipedal locomotion adaptation for stepping on obstacles using a novel foot sensor**, in *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, IEEE, 2022.
- [9] K. Sinaei and M. R. S. Yazdi, **Pid controller tuning with deep reinforcement learning policy gradient methods**, in *International Conference of Iranian Society of Mechanical Engineers (ISME)*, Tehran, Iran, 2021.
- [10] A. Vedadi, K. Sinaei, P. Abdolhazehzad, S. S. Aboumasoudi, and A. Yousefi-Koma, **Bipedal locomotion optimization by exploitation of the full dynamics in dcm trajectory planning**, in *RSI International Conference on Robotics and Mechatronics (ICRoM)*, IEEE, 2021.

SKILLS

Languages	C++, Python, C, MATLAB, C#, Shell
Robotics	ROS 1 / ROS 2, MoveIt, Nav2, Pinocchio, NVIDIA Isaac Sim, MuJoCo, Gazebo, PyBullet
AI / ML	PyTorch, CUDA, Hugging Face, Transformers, LeRobot, LangGraph, OpenCV, YOLO, Weights & Biases
Methods	Nonlinear, Adaptive & Robust Control, Control Barrier Functions (CBF), Model Predictive Control (MPC), Model Predictive Path Integral (MPPI), Trajectory Optimization, Convex Optimization, Reinforcement Learning (RL), Vision-Language-Action (VLA), Flow Matching, SLAM, State Estimation, Whole-Body Control
Tools	Git, Docker, Kubernetes, AWS, Linux, CI/CD, Unity
CAD / CAE	OnShape, SolidWorks, ANSYS, CATIA, AutoCAD

SELECTED PROJECTS

- Safe Vision-Language-Action Policies via Barrier-Enhanced Flow Matching** PhD Thesis | 2025 – Present
- Modified the Flow Matching sampling procedure of **VLA** policies (SmolVLA, Pi-0) so that Control Barrier Function constraints are enforced during action generation, giving safety guarantees for learned manipulation policies. Validated on hardware.
- Semantic Safety Filters over 3D Scene Graphs** Chewy | 2026
- Safety filter for physical-AI-driven mobile manipulators that reads semantic structure from a **3D scene graph (3DSG)** and prunes irrelevant obstacles, shrinking the constraint set evaluated at each control step to keep the filter tractable in real time.
- Wheeled Biped Robot** Open Source · Undergraduate Capstone | 2021 – 2022
- Motion planning and control for a 6-DoF wheeled-legged robot; most-starred of my open-source repositories. Code: github.com/Kassra-sinaei/wheeled_biped_robot
- Manipulator Teleoperation from a Smartphone** Open Source | 2024
- ARCore Android application streaming 6-DoF pose over WebRTC/UDP to collect demonstration trajectories for robot learning. Code: github.com/Kassra-sinaei/ArTeleop
- Deep Reinforcement Learning for PID Tuning** Independent Research | 2021
- PPO and actor-critic agents for automatic PID gain optimization, with custom Gym environments and a reward-shaping and logging pipeline.

LEADERSHIP

Penn State Advanced Vehicle Team (AVT) – Graduate Student Advisor

State College, PA | 2026 – Present

- Technical advisor to the undergraduate team competing in the **EcoCAR Innovation Challenge**, the U.S. Department of Energy Advanced Vehicle Technology Competition series managed by **Argonne National Laboratory**.

TEACHING

Pennsylvania State University – Teaching Assistant

State College, PA | 2024 – 2026

- Introduction to Linear Control Systems (EE 380) Spring 2026
- Continuous-Time Linear Systems (EE 350) Fall 2025
- Introduction to Digital Control Systems (EE 482) Spring 2024

University of Tehran – Teaching Assistant

Tehran, Iran | 2020 – 2021

- AI and Machine Learning in Mechanical Engineering Fall 2021
- Electrical Circuits and Electrical Machines Fall 2020

AWARDS & HONORS

- **Melvin P. Bloom Memorial Graduate Fellowship**, Penn State Electrical Engineering 2024
- **Milton & Albertha Langdon Memorial Graduate Fellowship**, Penn State Electrical Engineering 2023
- **Dr. Arthur H. Waynick Graduate Scholarship in Electrical Engineering**, Penn State 2022
- **College of Engineering Graduate Fellowship**, Penn State 2022 – 2023
- **Governmental BSc Scholarship** – full tuition waiver, national merit-based award 2017 – 2021

CERTIFICATIONS

NVIDIA Deep Learning Institute – Fundamentals of Deep Learning • AWS Certified Cloud Practitioner • ROS Industrial Training • Advanced Programming with C++

LANGUAGES

English – Full Professional Proficiency (C2, CEFR) • Persian – Native • Spanish – Elementary Proficiency (A2, CEFR)